

Experimental Report 01

Initial Baseline and Sensor Exploration

Remaining Useful Life prediction from multivariate sensor time series

Dataset	Task	Early signal	Baseline result
NASA C-MAPSS FD001	RUL regression	sensor_11 trend	MAE ~24 vs ~46

Dataset and setup

- Initial experiments use NASA C-MAPSS FD001 as a controlled benchmark for turbofan engine degradation.
- The target is Remaining Useful Life (RUL), measured as the number of operating cycles before failure.
- FD001 is used as the baseline setting before extending the pipeline to more complex operating conditions.

Sensor exploration

Initial exploration indicated that sensor trajectories contain degradation-related patterns mixed with hidden initial wear, process noise, and measurement noise. One early observation was that sensor_11 showed a visible upward trend over operating cycles, making it an informative candidate signal for RUL prediction.

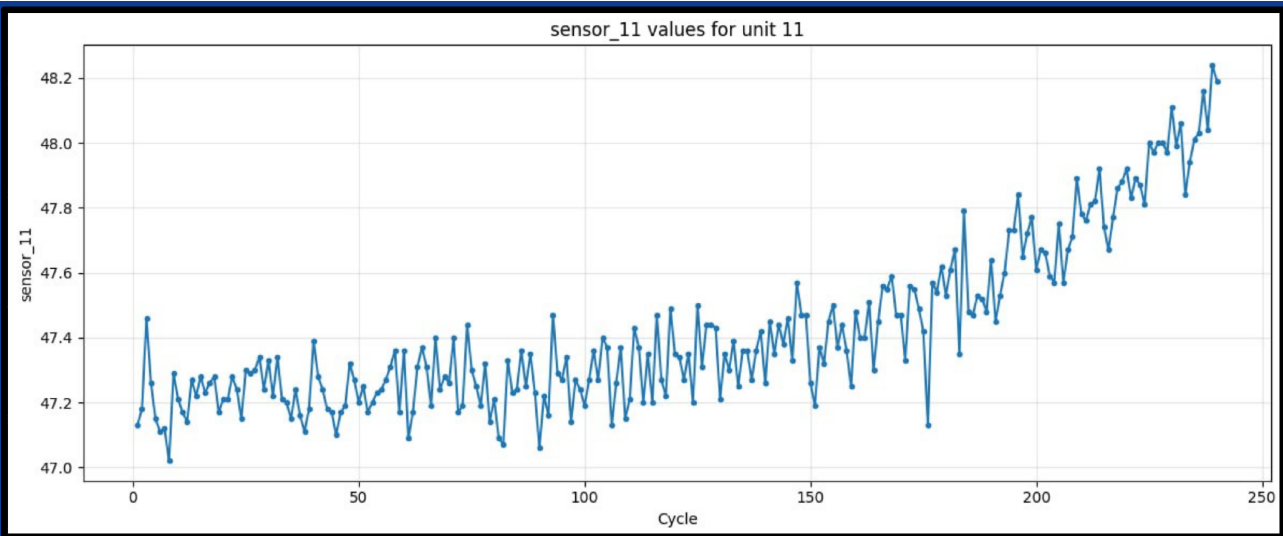


Figure 1. Example sensor_11 trajectory for one engine unit. The upward trend suggests that at least some sensor channels carry degradation information.

Baseline Result and Interpretation

Baseline prediction result

Approach	MAE, cycles
Naive baseline	~46
First baseline model	~24

MAE comparison, cycles



The first baseline model reduced the error by approximately 22 cycles, corresponding to about 48% lower MAE than the naive baseline.

Initial interpretation

- Even a basic model captures a measurable relationship between sensor readings and remaining useful life.
- The result supports further work on time-window representations, feature selection, and degradation-stage error analysis.
- The sensor exploration and baseline comparison provide a measurable starting point for the next experimental reports.

Report summary

Dataset	NASA C-MAPSS FD001
Main visual signal	sensor_11 upward trend over operating cycles
Naive baseline	MAE ~46 cycles
First baseline model	MAE ~24 cycles
Observed improvement	~22 cycles / ~48% lower MAE